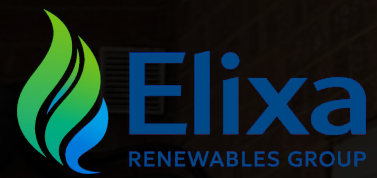


INSTALLATION CASE STUDY

Mitsubishi Ecodan | Full System Design and Upgrade

Clean Energy. Smarter Future.



The Heat Pump Is The Easy Part.

A modern house, sound fabric and every reason to expect a heat pump to perform. Whether it does comes down to the system designed around it, not the unit bolted to the wall.

SYSTEM

Mitsubishi Ecodan

REFRIGERANT

R32

SCOPE

Full system upgrade

INSTALLER

Elixia Renewables

THE BRIEF

A good house deserves more than a swapped box

This is a modern, well-built property with sound fabric and a heating system that worked. On paper, the easy job. In practice, the easy job is where most heat pump installations quietly go wrong.

The temptation on a house like this is obvious: take the old heat source out, put the new one in, connect it to whatever is already there and move on. It is faster, cheaper to quote, and wins the job against a properly designed alternative more often than it should.

It also produces the installations that give heat pumps their reputation: a unit forced to run at boiler flow temperatures because nobody checked the emitters, a cylinder too small for the household, pipework never sized for the lower temperature difference a heat pump works across, and controls left on whatever the factory shipped them with.

Smaller than people expect

The outdoor unit here is a single fan monobloc. It sits below the sill of the kitchen window, tight to the wall, on the existing patio. No compound, no screening, no rearranged garden. Mitsubishi's Ultra Quiet chassis covers 5 to 11.2 kW in this single fan format, which is enough for most modern family homes, and the current generation runs around three times quieter than the models that gave heat pumps their early reputation.

Size is the first question almost every customer asks. It is also the least important one. What decides whether this installation costs less to run than what it replaced is invisible from the driveway.

A heat pump is not a boiler that runs on electricity. Install it as though it were and it will cost more to run than the thing it replaced - and the customer will blame the technology.

The design envelope

Confirmed specification where stated. Remaining figures are indicative ranges typical of this class of property and system rather than measured values.

Heat pump	Mitsubishi Ecodan monobloc, R32 refrigerant
Property	Modern brick-built house, good existing fabric
Design heat loss	5-9 kW, typical of a property of this age and build
Design flow	35-45 degrees C, weather compensated
Hot water	Unvented cylinder sized to household demand
Scope	Full system: heat source, emitters, cylinder, controls

THE SOLUTION

Designed from the flow temperature outwards

Get the design flow temperature right and everything downstream follows. Get it wrong and no amount of good kit will rescue the running cost.

1 Heat loss calculated room by room

Not estimated from floor area, not carried over from the old boiler size. Every room assessed against its actual construction, glazing and exposure, because the whole design rests on that number being right.

2 Emitters checked against the target temperature

Each radiator was assessed for output at the design flow temperature rather than at boiler temperatures. Where a room fell short, that emitter was upgraded. Where it did not, it stayed - there is no merit in replacing radiators that already work.

3 Cylinder sized to the household, not to the cupboard

Hot water demand was sized against how the house is actually used. An undersized cylinder is the most common cause of a heat pump running hot and expensive all winter, and it is entirely avoidable at design stage.

4 Pipework sized and insulated for low temperature operation

Heat pumps move heat across a smaller temperature difference than boilers, so they need more flow to move the same energy. Undersized pipework throttles the system before it starts. External runs are fully lagged, as visible on the elevation.

5 Weather compensation set at commissioning

The controls were configured to the property, then the system was balanced and the curve adjusted on site. A default setting is not a commissioning.

The difference between a heat pump that costs less to run than the system it replaced and one that costs more is almost never the appliance. It is the twenty decisions made around it.

THE INSTALLATION

Neat where it shows, right where it does not

The outdoor unit sits on anti-vibration mounts on the existing patio, positioned for clear airflow and kept close to the point of entry so the primary runs stay short. External pipework is fully insulated and run in straight, level lines with valves accessible for service rather than buried behind the unit.

Indoors, the cylinder, controls and circulation were installed as one designed package. The system was flushed, dosed, balanced and commissioned against the design flow temperature, with the settings recorded and handed over rather than left for the customer to discover.

Scope of works

- Mitsubishi Ecodan monobloc air source heat pump, R32 refrigerant
- Room-by-room heat loss calculation and full system design
- Emitter assessment, with upgrades only where output required it
- Unvented hot water cylinder sized to household demand
- Primary pipework sized for low temperature operation and fully insulated
- Weather compensation controls configured and commissioned on site
- System flushing, inhibitor dosing, balancing and documented handover

Typical performance for a system of this type

3.5-4.2

Seasonal efficiency achievable on good fabric at a low design flow temperature

35-45 C

Design flow temperature, where the efficiency above comes from

5-9 kW

Typical design heat loss for a modern house of this size and build

Typical ranges for this class of property and system, confirmed per project at design stage. Efficiency is a consequence of design flow temperature and fabric, not of appliance brand.

If a quote does not tell you the design flow temperature, it is not a design. It is a price for a box on a wall.

Comparing heat pump quotes?

Ask every installer for the room-by-room heat loss, the design flow temperature and the cylinder sizing before you compare the price. The cheapest quote is usually the one that has not done them.

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